

WOODRUFF LAKELAND, WI, USA

WMO: 726404

Lat: 45.928N

Lon: 89.731W

Elev: 1630

StdP: 13.85

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 04865

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.9	-8.4	-21.5	1.8	-11.3	-16.6	2.4	-6.6	20.3	20.2	18.5	19.6	2.5	260	0.573

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	84.3	69.0	81.9	67.5	80.3	66.5	73.2	79.9	71.4	78.2	69.7	75.7	9.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
71.7	124.2	77.0	69.6	115.7	74.9	67.6	107.9	74.3	38.0	79.8	36.3	78.5	34.8	75.5	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.9	17.0	15.5	-21.1	90.1	3.6	3.1	-23.7	92.3	-25.8	94.1	-27.8	95.8	-30.4	98.1		
(5)			-21.3	76.3	3.5	2.7	-23.8	78.3	-25.9	79.9	-27.9	81.4	-30.4	83.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.2	14.8	16.5	27.9	40.7	53.8	63.0	67.2	65.1	58.2	45.2	32.6	19.7		
	DBStd	21.00	12.11	11.92	12.34	9.92	8.81	6.88	5.88	5.81	8.42	9.09	9.92	11.15		
	HDD50	4785	1090	937	692	308	59	2	0	1	22	208	526	940		
	HDD65	8611	1555	1357	1150	729	364	117	45	69	233	615	972	1405		
	CDD50	1937	0	0	7	29	175	390	534	470	268	60	4	0		
	CDD65	290	0	0	0	1	15	56	114	73	29	2	0	0		
	CDH74	2646	0	0	2	17	198	515	1056	611	229	18	0	0		
	CDH80	478	0	0	0	1	39	97	213	98	30	0	0	0		
	WSAvg	6.3	6.3	6.4	6.4	7.0	7.0	6.1	5.6	5.2	6.1	6.4	6.8	6.2		
	PrecAvg	31.4	1.1	0.9	1.7	2.5	3.5	3.9	3.7	4.2	3.7	2.8	2.1	1.3		
(15) Precipitation	PrecMax	42.4	3.3	2.9	4.7	5.4	7.9	10.8	6.9	13.8	7.2	7.3	5.9	2.6		
	PrecMin	16.1	0.1	0.0	0.2	0.3	0.4	1.4	1.1	0.4	0.5	0.2	0.4	0.2		
	PrecStd	5.5	0.7	0.6	1.1	1.3	1.7	1.7	1.6	2.3	1.9	1.5	1.3	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.5	51.7	69.0	76.9	84.7	86.4	90.2	86.1	82.5	76.9	63.1	44.5		
		MCWB	36.6	44.3	57.7	58.6	64.5	69.8	73.7	70.2	69.1	61.5	53.9	39.9		
	2%	DB	37.3	42.9	57.4	69.8	79.6	82.1	84.4	82.2	80.2	71.6	57.0	39.3		
		MCWB	34.5	38.2	45.8	54.0	62.8	67.3	70.1	67.7	67.4	61.1	50.1	36.6		
	5%	DB	35.6	36.9	51.8	62.8	75.0	79.9	81.9	80.5	76.8	64.1	52.0	37.0		
		MCWB	33.9	33.0	44.6	48.6	59.3	65.2	68.4	66.7	65.2	55.5	46.1	34.7		
	10%	DB	32.0	32.4	44.5	57.4	71.5	76.9	80.3	77.2	72.3	59.4	46.1	34.2		
		MCWB	30.7	30.2	38.8	45.6	58.4	63.8	67.3	66.0	63.7	52.4	41.6	33.1		
	0.4%	WB	37.0	45.1	58.5	60.4	70.0	74.2	77.4	75.3	73.2	67.5	56.1	41.7		
		MCDB	40.6	48.4	64.9	71.6	78.6	80.9	82.3	81.0	79.5	71.9	61.0	42.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.4	38.0	51.1	55.5	66.3	70.8	73.6	72.3	70.4	61.1	51.0	37.2		
		MCDB	37.8	42.5	54.2	65.3	74.8	78.6	80.4	78.3	75.3	67.7	55.6	38.9		
	5%	WB	32.9	33.1	44.2	51.0	62.8	68.6	71.7	70.5	68.2	57.5	46.1	35.3		
		MCDB	34.8	36.1	50.5	59.0	70.4	75.9	79.0	76.1	74.1	64.3	50.4	37.3		
	10%	WB	30.4	30.7	39.4	47.2	59.5	66.5	69.8	68.7	65.0	54.1	42.4	32.8		
		MCDB	31.6	32.4	44.7	57.0	67.2	74.1	76.0	74.7	71.7	59.4	46.3	34.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.2	18.1	19.1	19.7	21.7	20.5	20.8	20.1	19.5	16.1	13.0	12.9		
		MCDBR	15.9	19.1	23.5	29.4	26.8	23.7	22.8	23.0	23.0	23.0	20.9	13.1		
		MCWBR	13.7	14.6	15.1	16.5	13.3	12.1	11.5	11.7	12.6	14.0	14.8	11.3		
		MCDBR	14.9	16.3	20.4	25.2	21.1	20.1	19.8	18.5	18.3	19.7	18.6	11.2		
	5% WB	MCWBR	13.2	13.2	14.1	16.7	12.5	12.4	11.8	11.7	12.7	13.6	14.5	10.2		
(40) Clear-Sky Solar Irradiance	taub		0.239	0.264	0.306	0.330	0.360	0.359	0.359	0.358	0.341	0.305	0.280	0.249		
	taud		2.385	2.310	2.350	2.436	2.397	2.419	2.442	2.450	2.498	2.582	2.539	2.429		
	Ebn at Noon		282	293	294	295	288	287	286	282	279	277	265	266		
	Edh at Noon		22	30	33	34	36	36	35	33	29	23	20	19		
(44) All-Sky Solar Radiation	RadAvg		455	792	1183	1445	1702	1844	1912	1609	1239	761	443	322		
	RadStd		45	82	111	146	138	143	92	104	77	84	49	36		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page